

- 5 (a) Smooth muscle and cartilage are two of the tissues found in the walls of structures of the gas exchange system of mammals.

Complete Fig. 5.1 to show the distribution of these tissues in the gas exchange system of mammals.

Choose from the four structures listed below.

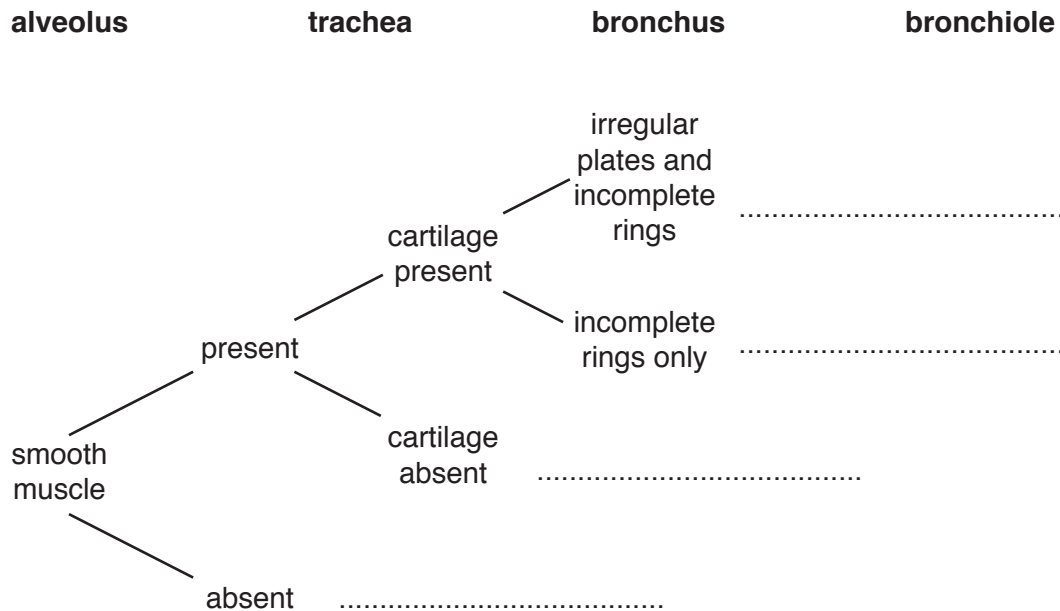


Fig. 5.1

[3]

Tobacco smoke is known to be one of the causes of lung cancer and chronic obstructive pulmonary disease (COPD).

- (b) Outline how tobacco smoke may cause lung cancer.

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..... [2]

- (c) Fig. 5.2 shows a section through the wall of one part of the gas exchange system in a person with COPD.

The tissue in the section of the wall labelled **X** is the result of changes to the original healthy tissue lining the lumen of the gas exchange system. The tissue shown is **not** scar tissue and is **not** a tumour.

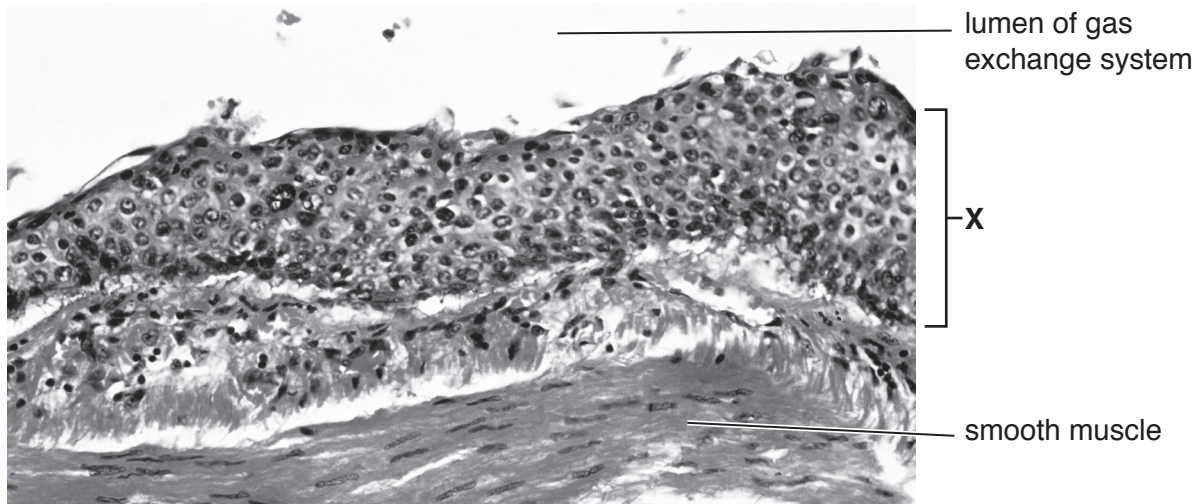


Fig. 5.2

The area labelled **X** on Fig. 5.2 is different in appearance to the original healthy tissue in the same part of the gas exchange system.

Describe these differences.

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[Total: 8]